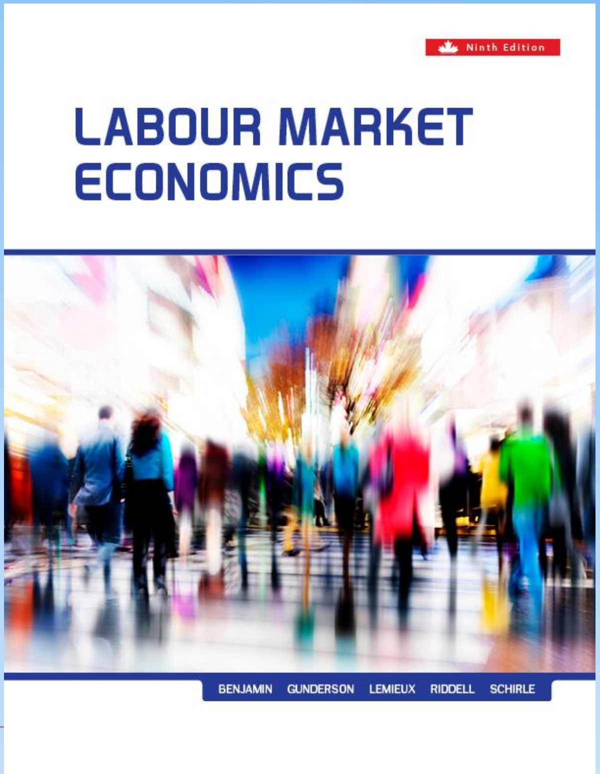
Chapter Three

Labour Supply and Public Policy: Work Incentive Effects of Alternative Income Maintenance Programs



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Learning Objectives

1. Discuss key features of different types of income maintenance programs, including demogrants, social assistance, unemployment insurance, wage subsidies, worker's compensation, and childcare subsidies.
2. Using the labour supply model, analyze and compare the work incentive effects of demogrants, social assistance programs, and wage subsidies.
3. Using the labour supply model, analyze the work incentive effects of unemployment insurance. Discuss key challenges to estimating effects of the program on individual behaviour.

Learning Objectives

4. Graphically illustrate the effect of a disability on the budget constraint. Discuss how well-being can be compared across disability compensation programs.
5. Illustrate how child care requirements affect the budget constraint and how programs to subsidize care can affect labour supply decisions.

Main Questions

- Can the myriad government transfer programs in Canada be analyzed in a common framework? If so, is there a way to determine the "best" type of income transfer program, at least in terms of its effects on work incentives?
- Does unemployment insurance unambiguously reduce incentives to work, thereby reducing overall employment?
- Can welfare programs be designed in a way that minimizes adverse work incentives, directing benefits to those who most need it?
- How can we incorporate disabilities into the labour supply model? Can workers' compensation reduce the incentive of injured workers to return to work?
- How do childcare subsidies encourage the labour force participation of women? How can childcare costs be incorporated into the labour supply model?

Income Maintenance Programs

- Designed to supplement low incomes, which can arise from a various number of reasons
- No single program can address the multiple reasons for low income

Income Maintenance Programs

Table 3.1 Government transfers received by Census Families in Canada, 2017		
Benefits:	Percent of families	Average Amount
All government transfers (federal and provincial)	83.7%	$12,750
Federal Child Benefits	20.2%	$6,314
Provincial Child Benefits	10.6%	$1,941
Employment Insurance	14.0%	$8,352
Social Assistance	11.3%	$8,367
Workers' Compensation	3.5%	$9,820
Canada/Quebec Pension Plan	32.4%	$10,092
Old Age Security	25.1%	$8,566
Guaranteed Income Supplement	9.6%	$6,634

Income Maintenance Programs

Universal Programs vs. Targeted Programs

Universal Programs

- Administratively simple
- Everyone receives the same transfer regardless of income
- Expensive
- Also benefits nonpoor

Income Maintenance Programs

Targeted Programs

- Cheaper method
- Individuals are given exactly enough of a transfer to reach the poverty line
- Only those below poverty line would receive transfer
- Creates a disincentive to earn income and, hence, may cause individuals to reduce work effort

Income Maintenance Programs

Permanent or Transitory?

- Compensating **low wages** or shortage of **hours worked**?
- Low income may be due to the shortage of **<u>hours</u>** worked or **<u>low wage</u>** rate:
  - Shortage of hours is mainly associated with the loss of transitory income (needs transitory type policy)
  - Low wage rate, on the other hand, may represent permanent low income (needs permanent type policy)

Difficult to isolate permanent from transitory

Demogrants, Welfare, and Wage Subsidies

Demogrants

Lump sum transfer or income grant (universal)

Specific to a demographic group

- Old Age Security (OAS): Providing monthly benefits to people aged 65 and over
- Former Universal Child Care Benefit (UCCB) program

Demogrants, Welfare, and Wage Subsidies

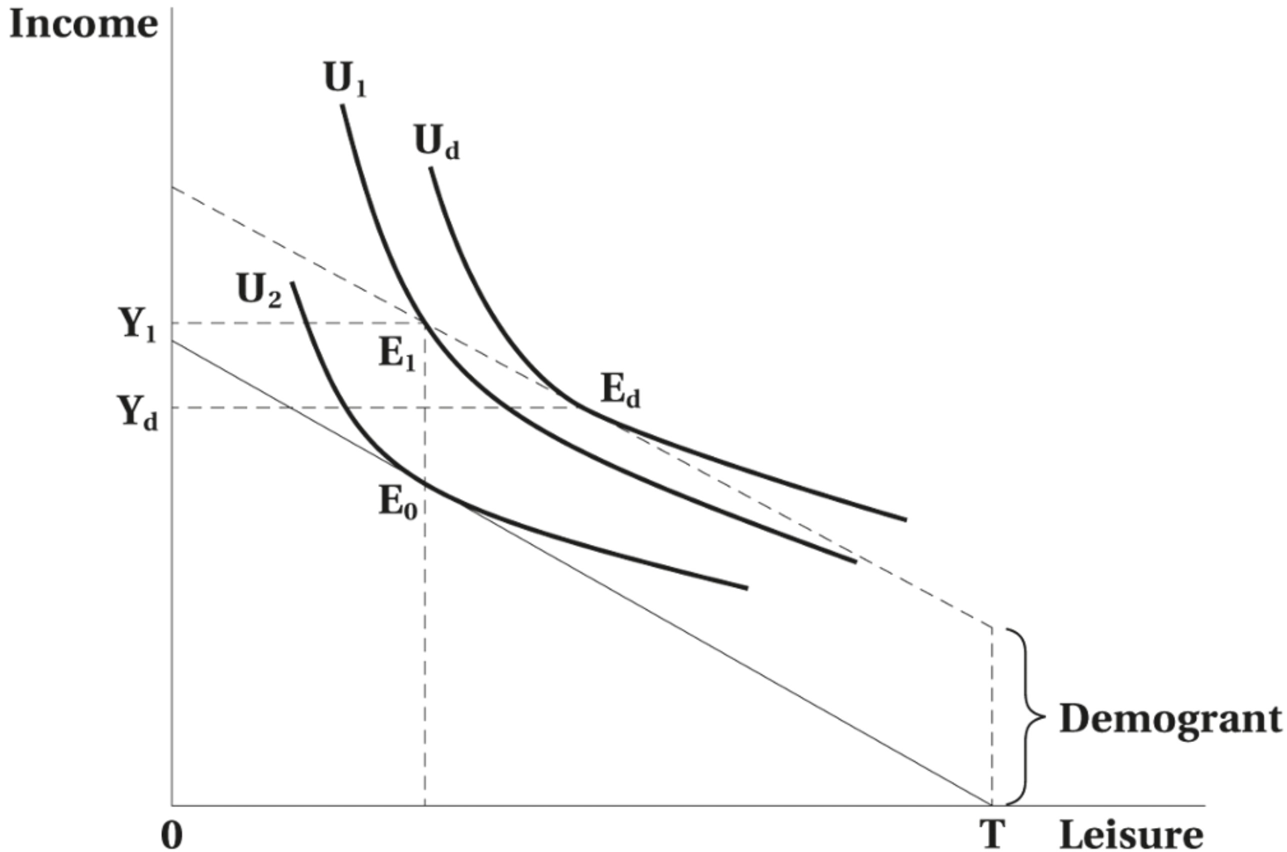


Figure 3.1 Work Incentive Effects of a Lump-Sum Demogrant

Demogrants, Welfare, and Wage Subsidies

Work Incentive Effects of a Lump-Sum Demogrant

- No substitution effect
- Work incentives are reduced
- Pure leisure-inducing income effect
- Increase in income is less than the amount of demogrant (all or a portion of the demogrant is used to buy leisure)

Demogrants, Welfare, and Wage Subsidies

Welfare

- Administered by the provinces
- Financed partly by the federal government
- Benefits depend on:
  - Needs of the family
  - Assets
  - Other sources of income

Demogrants, Welfare, and Wage Subsidies

Table 3.2 Income of Welfare Recipients by Province, 2018 (lone parent, one child)

Province	Annual Income[a] ($)	Income as % of Poverty Line[b]
Newfoundland	23,436	85
Prince Edward Island	20,977	77
Nova Scotia	18,240	67
New Brunswick	19,978	78
Quebec	21,867	86
Ontario	21,463	72
Manitoba	21,764	82
Saskatchewan	21,087	77
Alberta	19,927	68
British Columbia	20,782	71
Average (unweighted)	20,952	76

Demogrants, Welfare, and Wage Subsidies

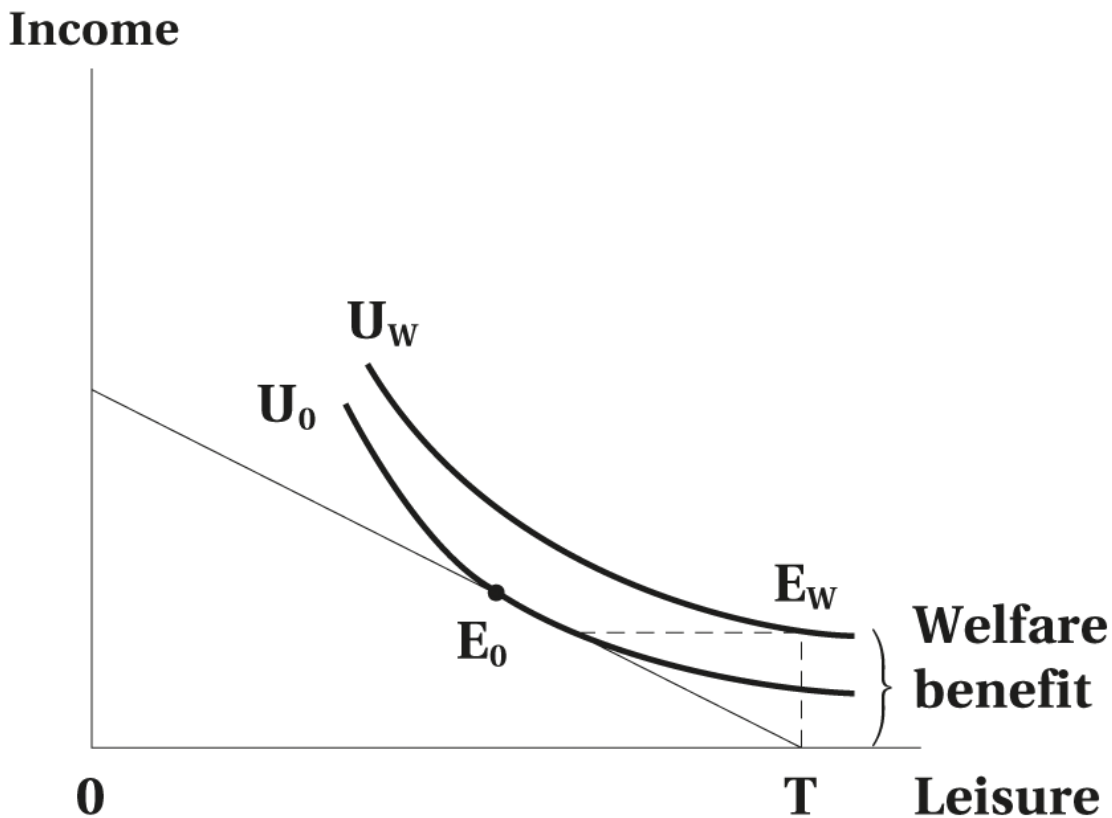


Figure 3.2 Work Incentive Effects of a Welfare Benefit with 100 Percent "Clawback"

Demogrants, Welfare, and Wage Subsidies

Work Incentive Effects of a Welfare Benefit with 100 Percent "Clawback"

- Adverse effect on work incentives
- Work is not chosen because of the 100% tax on earned income

Demogrants, Welfare, and Wage Subsidies

Possible Solutions to improve work incentive of welfare programs:

a) Reduce the welfare benefit
b) Increase the wage rate
c) Reduce the implicit tax (clawback)
d) Change the preference

Demogrants, Welfare, and Wage Subsidies

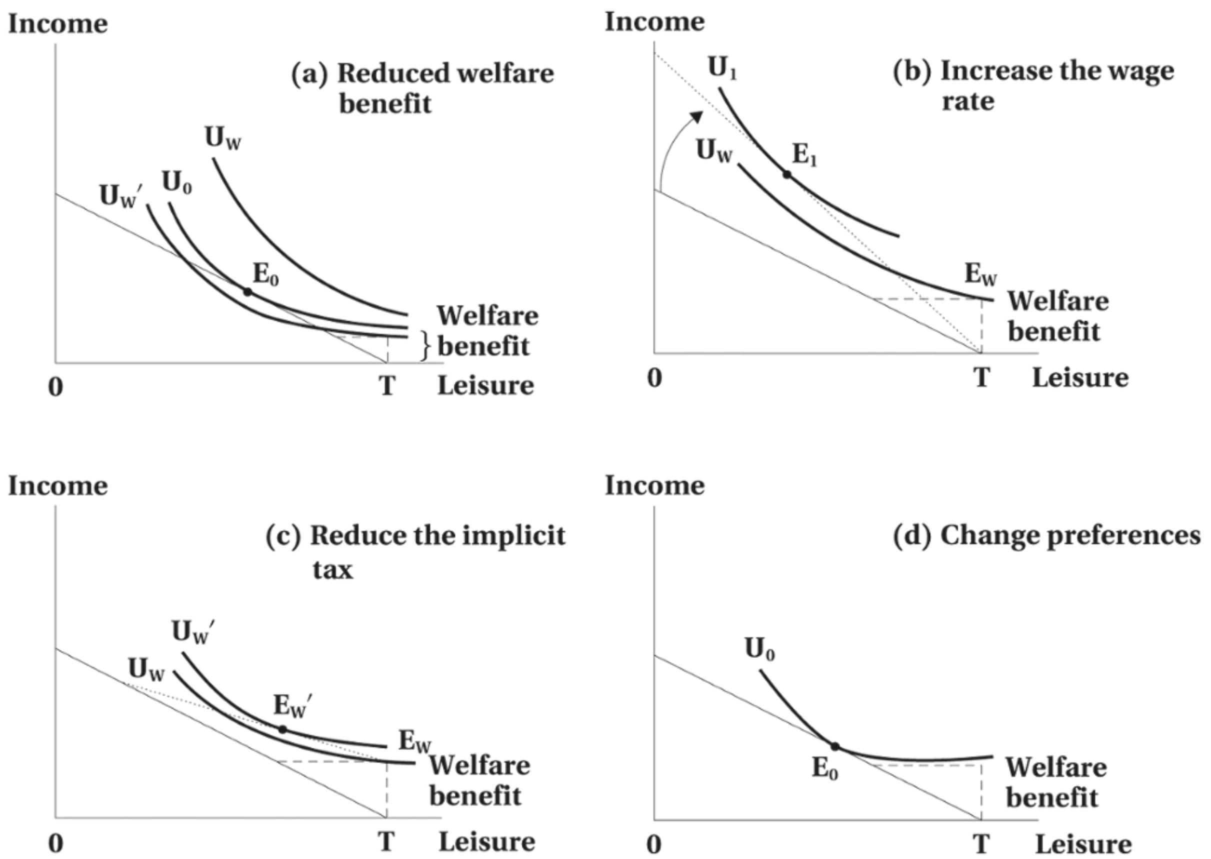


Figure 3.3 Other Work Incentive Effects of Welfare Programs

Demogrants, Welfare, and Wage Subsidies

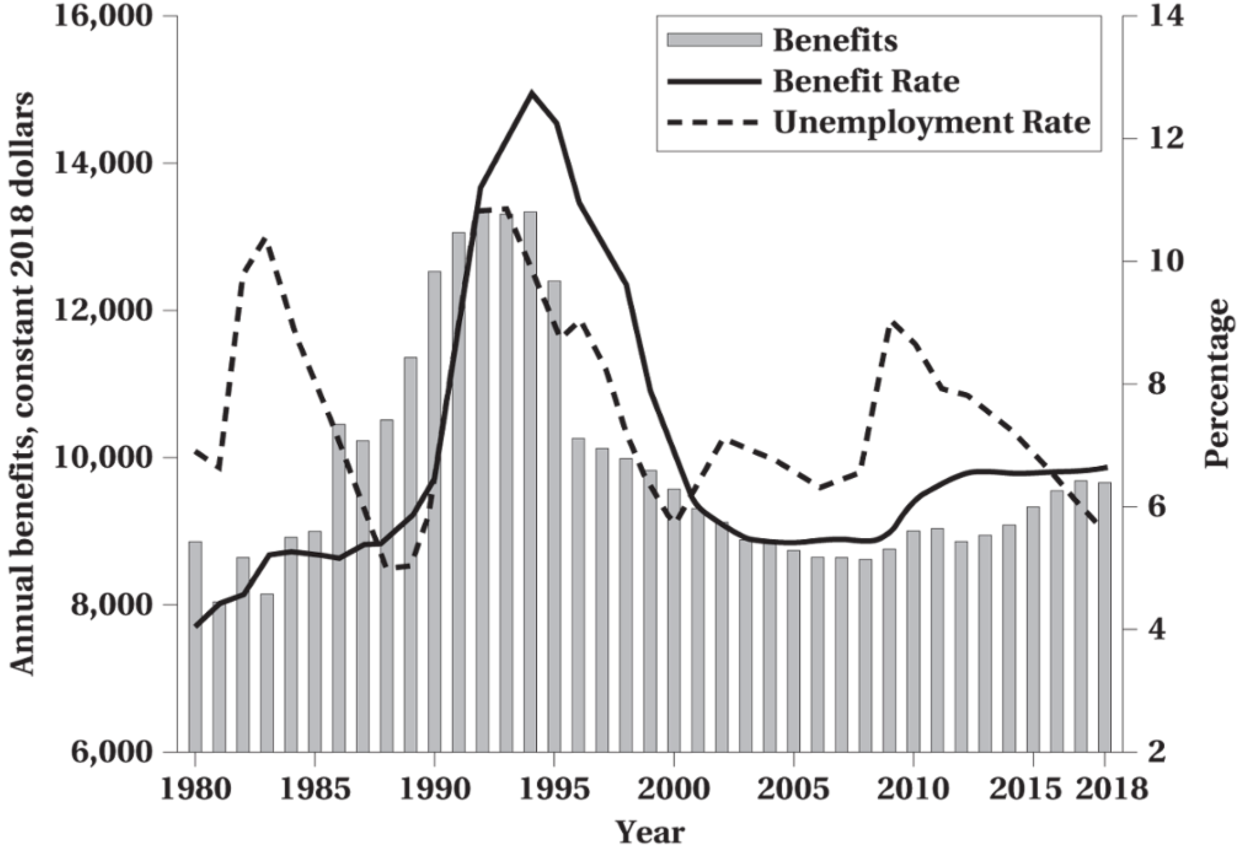


Figure 3.4 Welfare Benefits, Beneficiaries, and Labour Market Conditions, Ontario

Demogrants, Welfare, and Wage Subsidies

Negative Income Tax

- Income guarantee
- Implicit tax rate of less than 100%
- Recipients receive more from the guarantee than they will pay out in taxes
  - Child Tax Credit
  - Guaranteed Income Supplement

$$Y = G + (1 - t)E$$

Where: G = Guaranteed Income

t = Implicit Tax Rate

E = Labor Market Earnings

Demogrants, Welfare, and Wage Subsidies

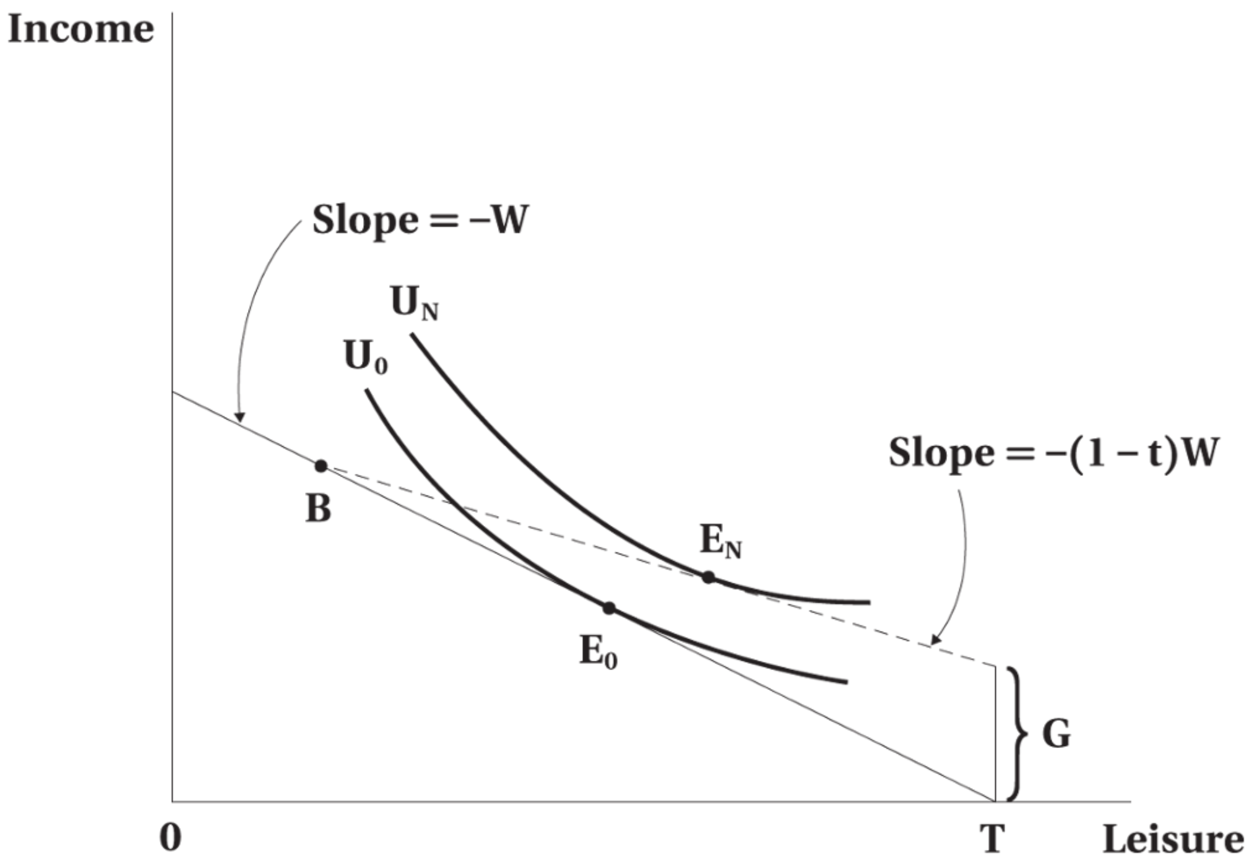


Figure 3.5 Work Incentive Effects of a Negative Income Tax

Demogrants, Welfare, and Wage Subsidies

Work Incentive Effects of a Negative Income Tax

- Work incentives will be reduced.
- The reduction in labour supply may have other desirable side effects.
- It may be used to replace some welfare programs that have even larger adverse incentive effects.

Demogrants, Welfare, and Wage Subsidies

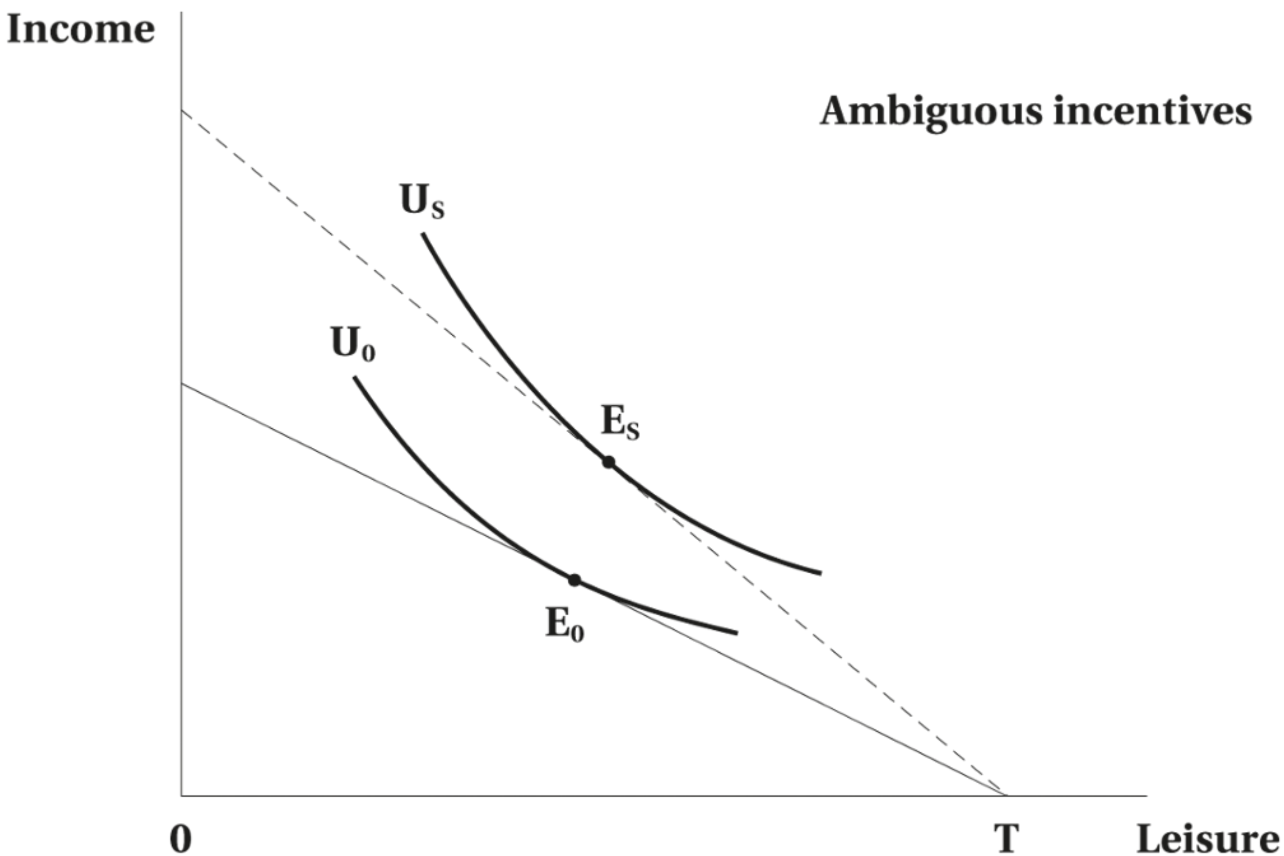


Figure 3.6 Work Incentive Effects of a Wage Subsidy

Demogrants, Welfare, and Wage Subsidies

Work Incentive Effects of a Wage Subsidy

- Theoretically indeterminate
- Adverse effects of wage subsidy are not as great as those of the negative income tax
- Disadvantage: does nothing for the income of those who are unable to work

Demogrants, Welfare, and Wage Subsidies

Refundable Tax Credit

- A new "income-tested" wage subsidy program used in both US and Canada to reduce the cost associated with the pure wage subsidy programs
- It is a refundable tax credit or subsidy that is paid irrespective of other income taxes paid by the individuals

Demogrants, Welfare, and Wage Subsidies

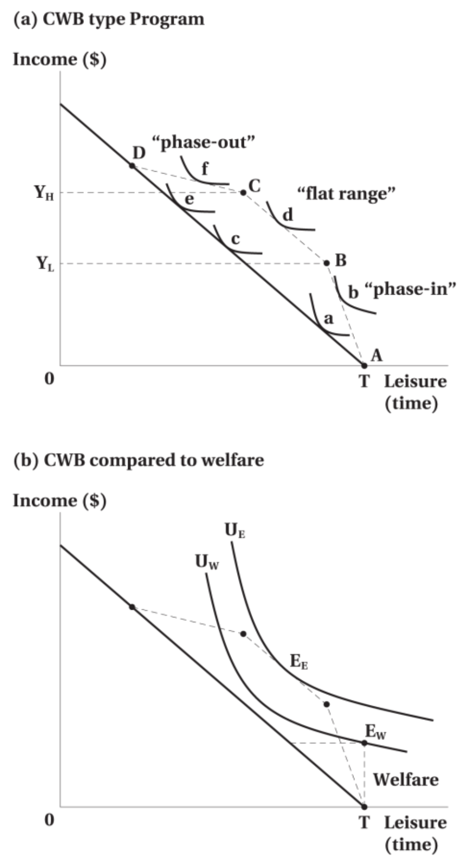


Figure 3.7 Work Incentives of a Refundable Tax Credit

Demogrants, Welfare, and Wage Subsidies

Work Incentives of a Refundable Tax Credit

- The **phase-in** section is similar to a standard wage subsidy. The substitution effect increases work effort while the income effect reduces work effort. The overall impact on work effort is ambiguous.
- In the **flat range**, the program exerts only an income effect and labour supply is reduced, assuming leisure is a normal good.
- The **phase-out** section is like a negative income tax. The substitution and income effect reduce work effort. While the net effects on hours worked may be ambiguous, the effect on participation is clearer.

Unemployment Insurance

- Canada's unemployment insurance program, known as *Employment Insurance* (EI) since 1996, is the largest income security program for non-elderly individuals in the country.
- The amount of income replacement rate is 55% of lost earning, subject to a maximum.
- The duration of benefit ranges from 14 to 45 weeks, depending on the regional rate of unemployment.
- In order to qualify for the benefits, individuals must have worked between 420 and 700 hours, the equivalent of 12–20 weeks at 35 hours per week, again depending on the unemployment rate of the region.

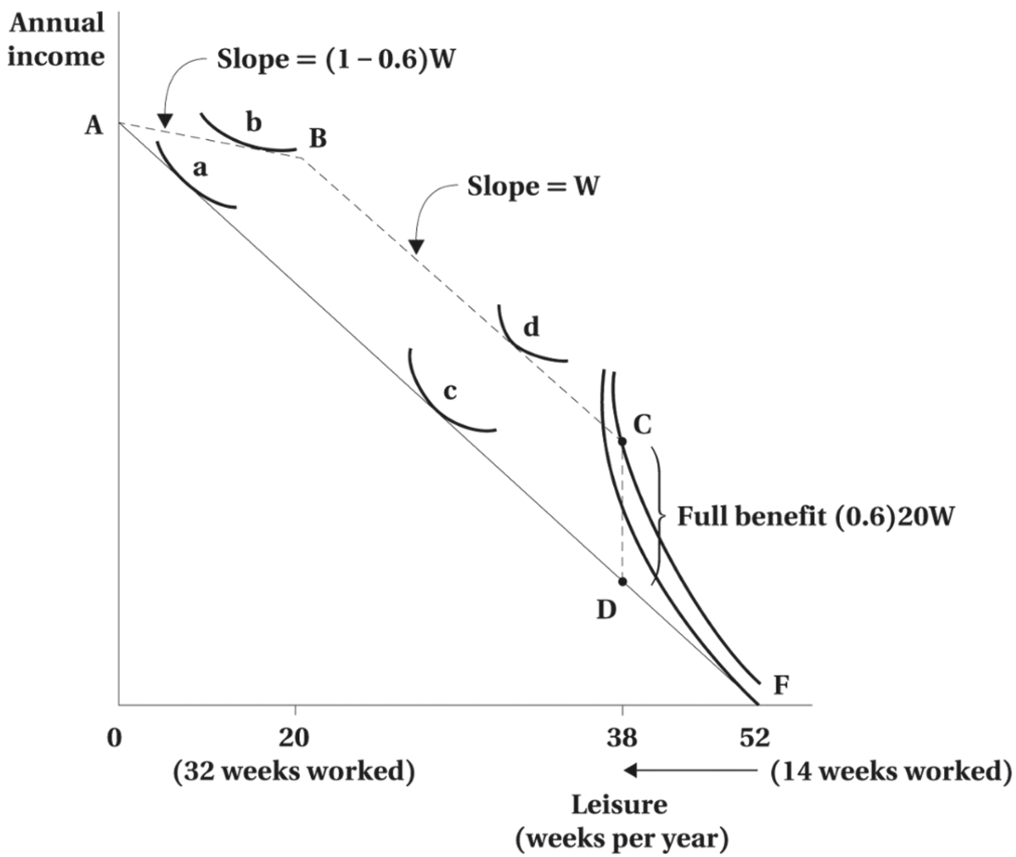
Unemployment Insurance

Worked Example

Unemployment Insurance and Work Incentives

- Recipients receive 60 percent of their weekly pay in the case of unemployment
- It requires a minimum of 14 weeks of employment in order to qualify for benefits.
- The maximum period of receiving benefits is 20 weeks

Unemployment Insurance



Income Constraint for a Simplified Unemployment Insurance Scheme

Disability and Workers' Compensation

Effect of a Disability

- Budget constraint or preference curve could be altered
- Factors to be considered:
  - hours able to work
  - medical expenses
  - reduced ability to earn wages
  - disutility of labour market vs. other activities

Disability and Workers' Compensation

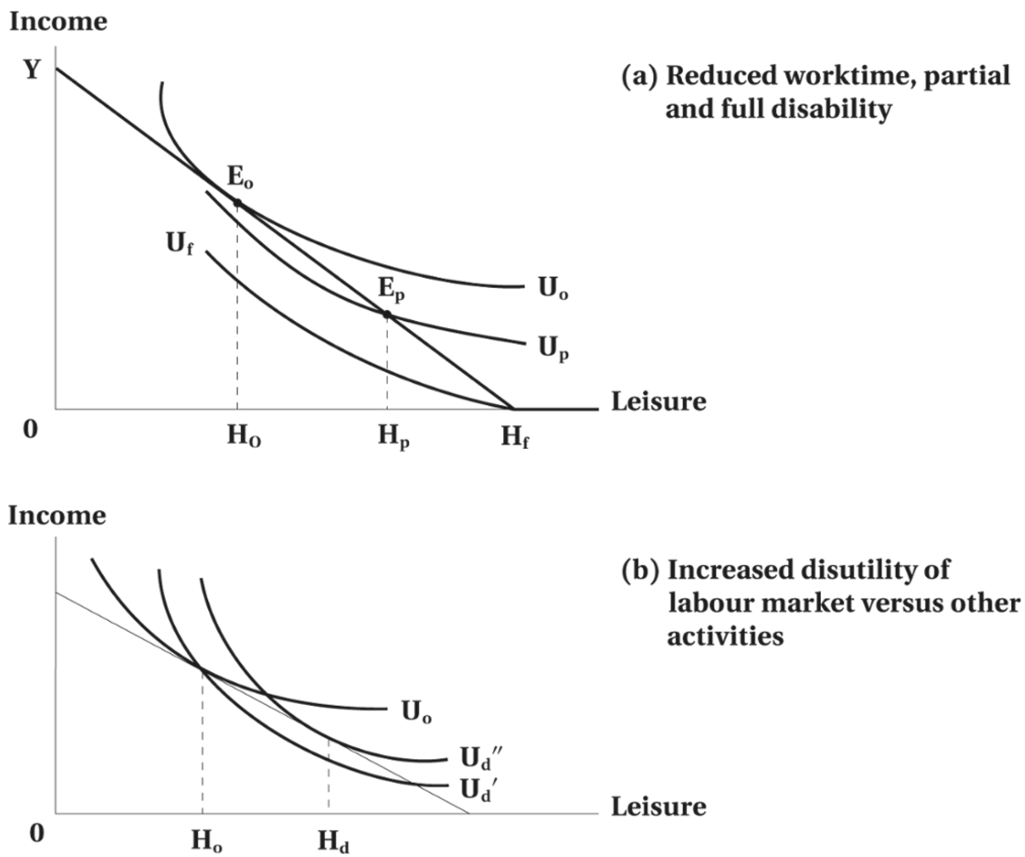


Figure 3.8 Effect of Disability

Disability and Workers' Compensation

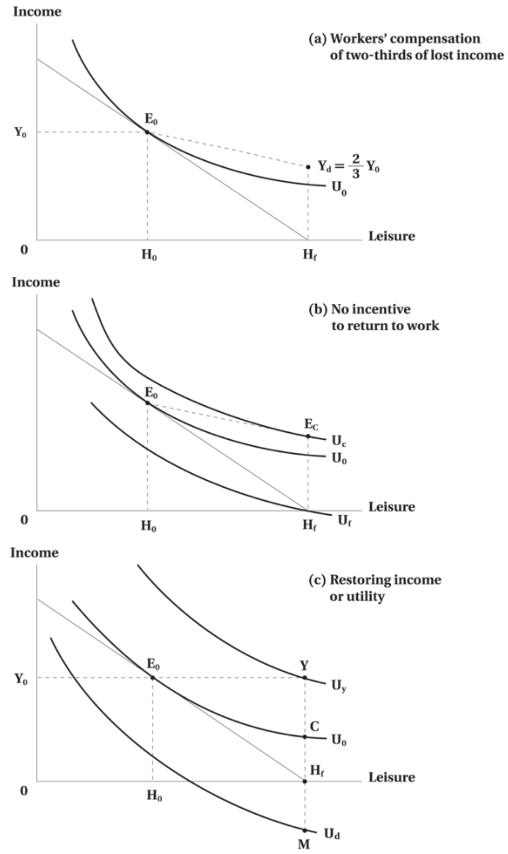


Figure 3.9 Effect of Compensation

Child Care Subsidy

- Daycare costs can be modelled as a special case of **fixed costs** associated with working
- Effect of childcare subsidy
  - Fixed costs may increase reservation wage
  - Fixed costs may also affect hours-of-work
    - Increased hours for those who participate
    - Create a discontinuity in labour supply

Child Care Subsidy

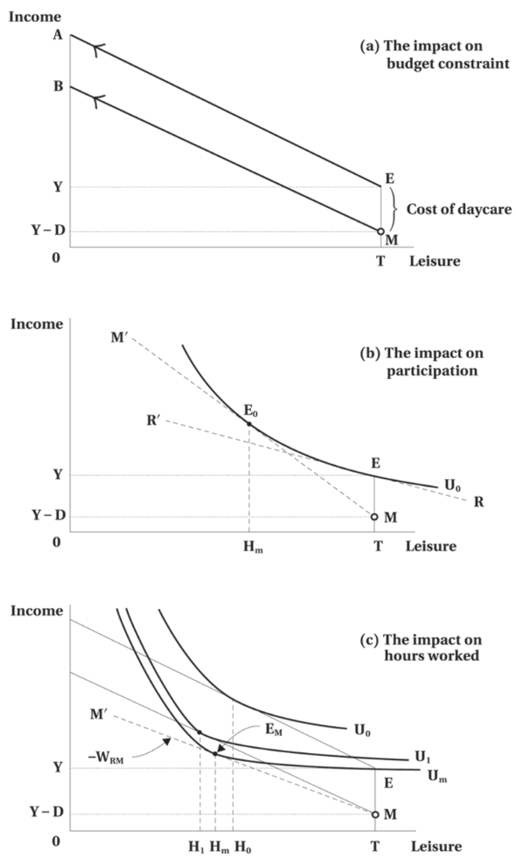


Figure 3.10 The Effect of Child Care Costs on Labour Supply

Child Care Subsidy

- Encourages labour force participation and part-time work
- Reduces the hours of work for those already participating

Chapter Summary

- Income maintenance schemes are designed to top-up low incomes, which can arise for any of a variety of reasons, be permanent or transitory, and may be due either to low wages or hours.
- The labour supply framework is a useful way to analyze work incentive effects.
- Demogrants are pure lump-sum transfers, based only on immutable individual characteristics like age or sex.
- Welfare programs are designed to increase the income of individuals with low income. The degree to which they have adverse work incentive effects depends on how strictly benefits are reduced in response to higher earnings.

Chapter Summary

- Negative income taxes, or guaranteed annual income programs, are generalizations of welfare programs.
- Wage subsidies make work more attractive but are very expensive unless they are targeted to the working poor. One increasingly popular form of targeted work subsidy is delivered as a refundable tax credit.
- Unemployment insurance is designed as an insurance scheme responding to short-run reductions in available hours of work, especially due to cyclical changes in labour market conditions.
- Workers' compensation is designed to address income losses arising from workplace injuries.
- Childcare expenses can be modelled as fixed costs associated with working.